# Koa Shen

Mechanical Engineering | Supplier Industrialization, DFM & Manufacturing Quality | Summer 2027

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## EDUCATION

**University of California, Santa Barbara** Expected June 2028

*B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94*

- Dean's Honors every quarter. Relevant coursework: Mechanics of Materials, Dynamics, Circuits, MATLAB.

## MANUFACTURING & ENGINEERING EXPERIENCE

**Mechanical Engineering Intern** Summer 2026 - Present

*SLAC National Accelerator Laboratory, LCLS | Menlo Park, CA; extended part-time remote*

- Produced Solid Edge designs and PDM-controlled drawings with GD&T for manufacture and inspection of beamline motion assemblies.
- Redesigned mounts, alignment features, and detector strain relief; checked load cases to address crash and drop risks around $100k+ detectors.
- Built a Python/Drake digital twin for a 30-DOF, 305-body assembly; millisecond collision checks identified at least five physical pinch points and informed design changes.
- Developed 15 reusable motion-stage models; the framework is being adopted across three LCLS assemblies.

**Fabrication Team Lead** Fall 2024 - Present

*Gaucho Racing, Formula SAE Electric | UC Santa Barbara*

- CNC-machined 33 precision suspension components from 4130 steel and 7075-T6 aluminum on Haas mills and lathes; held clevis and rod-end bores within a 0.0004 in tolerance window and bearing interfaces within 0.0005 in.
- Measured critical fits with go/no-go tests and worked directly with designers on GD&T and manufacturability; proposed standardized spindle geometry to reuse proven CAM programs and simplify setup.
- Led 15 teammates through driver safety-system design, manufacturing, and installation; GR26 passed FSAE technical inspection fully rules-compliant.
- Own chassis and ergonomics packaging, coordinating welded structures, assembly interfaces, and manufacturing across vehicle subteams.

**Maintenance Technician** Summer 2025

*Exploratorium | San Francisco, CA*

- Serviced 100+ public exhibits and completed three major overhauls; redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement-part costs.
- Reverse-engineered components and developed CAD, CAM, and fixturing for replacement parts; documented remanufacturing instructions so technicians could reproduce repairs.

## TECHNICAL SKILLS

**CAD & Quality:** SolidWorks, Solid Edge, Teamcenter PDM, GD&T, drawing interpretation, go/no-go inspection, design documentation

**Manufacturing:** Haas CNC mills and lathes, Mastercam/CAM, manual machining, welding, 3D printing, fixturing, assembly, DFM

**Analysis & Tools:** Python, MATLAB, FEA, statics/dynamics, kinematics, Drake, Git